When Marginal Coverage Hides the Boundary: Evaluating Conformal Regions for Compositional Responses

Autonomous research study

Abstract

Conformal regions for a composition must cover a whole vector, including its possible zero components. We study how evaluations based on marginal coverage and interior volume can obscure failures on the boundary of the simplex. We give an exact finite-calibration formula for the whole-simplex transition of an unsmoothed centered log-ratio score extended to infinity at zeros, and evaluate distance, density, coordinatewise, scaled, weighted, and stratified conformal constructions in controlled Dirichlet experiments. Across independent repetitions, pooled methods generally attain nominal marginal coverage under exchangeability but can severely under-cover high-dispersion groups and structural-zero responses. In one mixed three-component setting, pooled square-root distance regions cover boundary responses at 0.187 ± 0.011; a candidate-response and covariate stratification control reaches 0.921 ± 0.007, with substantially larger regions. Smoothing restores finite log-ratio scores but does not repair boundary coverage. A true-density oracle is efficient under a specified reference measure yet can also sacrifice a minority group. We recommend reporting coverage by relevant covariate and support strata together with stratum-specific size, and show exactly which guarantees each calibration choice provides. Results are synthetic and concern observed compositions, not latent probabilities inferred from single class labels.

1 Introduction

Probability vectors, topic proportions, and mixture fractions live on a closed simplex. A point estimate in this space does not specify a useful uncertainty region. Conformal prediction can attach a set with finite-sample marginal coverage, but that statement leaves substantial freedom in the shape, support, and allocation of errors across examples. A method can cover most compositions while systematically missing those with a zero component, and still satisfy its advertised guarantee.

We distinguish two problems that share a probability-vector predictor. In classification, the future response is a label and the conformal output is a subset of labels. Here the future response is an *observed composition* $Y \in \Delta^{K-1} = \{y \in [0,1]^K : \sum_j y_j = 1\}$. The output is a subset of this continuum and the coverage event is simultaneous inclusion of all coordinates. The softmax vector of a classifier is not itself a ground-truth composition. Without probability-valued targets or further observation assumptions, label coverage cannot establish coverage of a latent probability vector.

The familiar distinction between marginal and conditional coverage becomes especially consequential when the response distribution charges different faces of the simplex. Interior volume assigns every proper face size zero, regardless of its probability mass. Log-ratio scores create an additional difficulty: they are undefined at zeros. Common repairs include smoothing, but the impact of that repair on set size and support-specific reliability must be evaluated on the original target scale.

We ask four concrete questions. (Q1) At matched nominal marginal coverage, how do geometric scores differ in efficiency? (Q2) What exactly happens to raw log-ratio calibration as zero prevalence

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crosses the allowed error rate? (Q3) Which observable-group and candidate-response stratifications repair which coverage failures, and at what size cost? (Q4) Which conclusions survive a controlled change in group prevalence? Our hypotheses and original design were recorded in `PROTOCOL.md` before the main experiment. Two response-stratified controls were added after inspecting the initial results; we identify that analysis as exploratory.

Our contribution is a focused evaluation and an exact analysis of a boundary failure mechanism, not a new conformal algorithm. We provide a reproducible experiment with common predictions and paired randomness, an interior/face size diagnostic, and an oracle baseline under an explicit reference measure. We find that the best choice depends on the target guarantee and loss: marginal size optimization, group reliability, and boundary reliability are different objectives.

2 Related work and scope

Split conformal inference separates fitting from calibration and permits arbitrary fixed score functions. Its marginal guarantee and its difference from full conformal inference are established in Lei et al. [2016]. Angelopoulos and Bates [2021] explicitly recommend checking adaptivity and subgroup coverage and describe scale normalization and group-balanced calibration. We use these constructions as baselines, rather than claiming that the need for conditional diagnostics is new. Full conformal and resampling methods principally address data reuse and refitting; comparing that tradeoff is outside this experiment, which holds the predictor fixed to isolate geometry and calibration.

Class-label sets such as RAPS [Angelopoulos et al., 2020] target a discrete response; their cardinality is not the volume of a region for a probability-valued response. More directly, Javanmardi et al. [2024] construct conformal credal sets from probability-valued labels using distance and likelihood scores, investigate annotation noise, and evaluate set inefficiency. Thus conformalizing the simplex and using a density score are prior work. Our narrower distinction is between support-specific reliability and geometric size when exact zeros have positive probability, including a finite-calibration saturation formula and stratified efficiency measurements. Our regions need not be convex, so we use the term *region* rather than imposing a convex credal-set interpretation.

Covariate shift breaks ordinary exchangeability. We implement the weighted split construction of Tibshirani et al. [2019], including the test-point atom at infinity, with known likelihood ratios. This is a controlled validity check, not an evaluation of estimated shift correction. To preserve independent research under the task's access constraint, we consulted only the five supplied starter papers cited here via direct PDF URLs; we made no thematic literature search. Our positioning is consequently relative to this restricted, explicitly documented reading set, not a comprehensive novelty claim.

3 Regions, guarantees, and evaluation

Split calibration. Conditional on a predictor $m(x)$ and any pilot estimates fitted independently of calibration, let $S_i = s(X_i, Y_i)$, $i = 1, \dots, n$. For miscoverage α, write

$$k = \lceil (n+1)(1-\alpha) \rceil, \quad \widehat{q} = S_{(k)}, \quad C(x) = \{y \in \Delta^{K-1} : s(x,y) \le \widehat{q}\}, \tag{1}$$

where $S_{(n+1)} = +\infty$. We use nonrandomized inclusion of ties. Exchangeability gives $\mathbb{P}\{Y_{n+1} \in C(X_{n+1})\} \ge 1-\alpha$, also for extended-real scores. Near-equality requires additional no-tie conditions; it need not hold with an atom at infinity. Coverage is averaged over calibration and test draws, not a guarantee for every realized calibration sample.

Geometric and coordinate scores. The main scores are $s_{\max} = \|y - m\|_\infty$, $s_2 = \|y - m\|_2$, and $s_H = \|\sqrt{y} - \sqrt{m}\|_2$, where square roots are componentwise. The last differs from a conventional Hellinger distance only by a fixed factor, which cancels under calibration. For positive vectors define $\mathrm{clr}(y)_j = \log y_j - K^{-1} \sum_\ell \log y_\ell$. Raw CLR uses $s_{\mathrm{CLR}} = \|\mathrm{clr}(y) - \mathrm{clr}(m)\|_2$ in the interior and $+\infty$ otherwise. In this explicitly chosen extension, $\widehat{q} = +\infty$ yields the entire *closed* simplex. This convention is necessary for validity of this construction; silently discarding boundary candidates would fail.

Smoothed CLR uses $\mathrm{clr}(y + \epsilon \mathbf{1})$ on both candidates and centers, with $\epsilon = 10^{-3}$ as the main setting and $10^{-4}, 10^{-2}$ as sensitivity controls. Dividing by $1 + K\epsilon$ would cancel inside CLR. Membership

always tests the original y; we do not replace the evaluation target by a smoothed vector. Every smoothed score is finite on the compact simplex, although its interpretation as a relative error changes with ϵ and K.

Coord calibrates $|Y_j - m_j(X)|$ separately at error α and intersects the resulting intervals with the simplex. It is a deliberately invalid control for *joint* $1-\alpha$ coverage. Bonf uses error α/K for each coordinate and therefore has a union-bound joint guarantee. In both cases we use the exact order statistic, including infinity for undersized calibration sets.

Adapting calibration. H-scale divides s_H by a pilot-estimated group RMS score a_g. H-group calibrates s_H separately within a prespecified covariate group G. The first keeps a marginal guarantee but makes no group guarantee; the second guarantees coverage conditional on each G under within-group exchangeability. H-weight uses the weighted empirical score distribution with masses proportional to $w(X_i) = dP_{\mathrm{test,X}}/dP_{\mathrm{cal,X}}(X_i)$ and an additional mass $w(x)$ at infinity [Tibshirani et al., 2019].

The exploratory controls H-Z and H-GZ use the response partition $Z(y) = \mathbf{1}\{\min_j y_j = 0\}$. With calibration thresholds q_z or $q_{g,z}$ they return

$$C_Z(x) = \{y : s_H(x,y) \le q_{Z(y)}\}, \qquad C_{GZ}(x) = \{y : s_H(x,y) \le q_{G(x),Z(y)}\}. \tag{2}$$

No knowledge of the realized test target is used: the candidate determines its stratum during membership evaluation. This is the standard response-conditional conformal construction applied to a fixed support partition. It guarantees coverage within Z, or within each (G, Z) cell, provided calibration and test examples are exchangeable within that cell. Small or empty cells can force infinite thresholds; this includes boundary strata absent from a finite calibration sample. Such a threshold includes only its candidate stratum, unlike the whole-simplex collapse of raw CLR. These partitions do not guarantee coverage within individual faces or at every x.

Reliability and size. We report marginal, G-conditional, and boundary/interior-conditional coverage. We also measure size relative to a declared reference distribution, not by raw score radius. For $K = 3$, let U_2 be the uniform distribution on the triangle and U_1 an equally weighted mixture of uniform distributions on its three edges. Define

$$V_r(C) = \mathbb{E}_{X_{\mathrm{test}}}\mathbb{E}_{U_r}[\mathbf{1}\{U_r \in C(X_{\mathrm{test}})\}], \quad V_\lambda = \lambda V_2 + (1-\lambda)V_1. \tag{3}$$

Both are dimensionless fractions. V_1 does not replace actual boundary coverage: it weights edges uniformly whereas the target law need not. We present (V_2, V_1) separately and use $V_{1/2}$ only when a scalar is needed. The weight is an evaluation preference, not a probability estimated from the data.

Proposition 1 (Volume blindness)**.** *For any measurable region C, removing all its boundary points leaves its $(K-1)$-dimensional volume unchanged but reduces coverage by $\mathbb{P}\{Y \in C(X) \cap \partial\Delta^{K-1}\}$. Thus interior volume alone cannot distinguish regions that differ in boundary reliability.*

The statement follows because the boundary is a finite union of lower-dimensional faces and hence has zero interior volume. It motivates reporting coverage and size by support stratum; it does not select a universal scalar efficiency criterion.

4 An exact boundary transition for raw CLR

The next result quantifies why a valid implementation can be very conservative rather than merely numerically unstable.

Proposition 2 (An atom at the maximum score)**.** *Let calibration and test scores be i.i.d. on $\mathbb{R}\cup\{+\infty\}$ with $\mathbb{P}(S = +\infty) = \pi$ and an atomless conditional law given $S < +\infty$. For $1 \le k \le n$ in Eq.* (1)*,*

$$\mathbb{P}(\widehat{q} = +\infty) = \mathbb{P}\{\mathrm{Bin}(n,\pi) > n-k\}, \tag{4}$$

$$\mathbb{P}(S_{n+1} \le \widehat{q}) = \frac{k}{n+1} + \left(1 - \frac{k}{n+1}\right)\mathbb{P}\{\mathrm{Bin}(n+1,\pi) > n+1-k\}. \tag{5}$$

For raw CLR with positive centers and scores finite on every interior target, the first event is exactly the whole-simplex event. Any finite-radius raw CLR region excludes every boundary target, so its population coverage cannot exceed $1-\pi$.

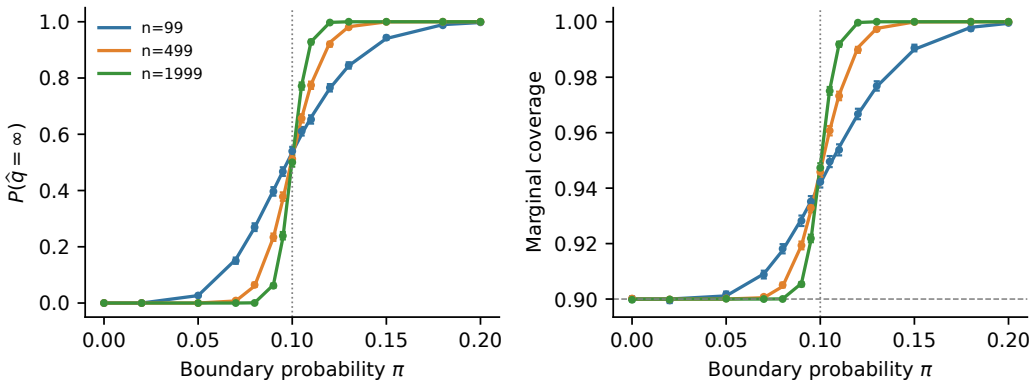


Figure 1: Exact atom transition: lines are Eqs. (4)–(5); points are 4,000 independent calibration arrays per setting. Left: 95% Wilson intervals for collapse probability. Right: coverage is Rao–Blackwellized over the test score, with approximate 95% Monte Carlo intervals. Finite scores are exponential, and infinite scores have probability π.

Proof sketch. The calibration quantile is infinite precisely when fewer than k calibration scores are finite, giving Eq. (4). Among $n+1$ scores, if at least k are finite, averaging the leave-one-out inclusion event covers exactly the k smallest finite scores. If fewer are finite, every leave-one-out threshold is infinite and all scores are covered. Exchangeability gives Eq. (5). A complete argument and the stratification guarantee are in Appendix A.

For fixed $\pi < \alpha$, collapse probability tends to zero as n grows; for $\pi > \alpha$ it tends to one. At $\pi = \alpha \in (0, 1)$ it tends to one half. More calibration data therefore sharpens this transition; it does not resolve it. When $\pi > \alpha$, obtaining a nontrivial region requires changing the score or its treatment of support. Smoothing is one option but supplies only a marginal guarantee. These statements concern the specified raw extension, not all log-ratio-based constructions.

5 Experimental design

Controlled compositions. We use entirely generated data, so exact support and noise strata are known. Independently draw $T \sim \mathrm{Unif}[-1, 1]$ and $G \sim \mathrm{Bernoulli}(.25)$, set $X = (T, G)$, and define $m(T)$ through a fixed softmax (Appendix B). Draw $Y^* \mid X \sim \mathrm{Dirichlet}(\kappa_G m(T))$. With probability π_G, choose one coordinate uniformly, set it to zero, and renormalize the remaining coordinates. Otherwise $Y = Y^*$. There are no vertices or multiple-zero targets. The same m is used by every distance method: it is the conditional mean before deletion, and need not be the mean afterward. This intentionally isolates region construction from predictor fitting, at the expense of realism.

The four exchangeable settings use $(\kappa_0, \kappa_1; \pi_0, \pi_1)$ equal to Dense $(40, 40; 0, 0)$, Heterogeneous $(160, 8; 0, 0)$, Zeros $(40, 40; .15, .15)$, and Mixed $(160, 8; .02, .32)$. Mixed has marginal boundary probability .095, just below $\alpha = .1$, with boundary mass concentrated in the high-dispersion group. Shift uses the same conditional laws as Mixed but changes only test $P(G = 1)$ to .65. Exact weights are $.65/.25$ for $G = 1$ and $.35/.75$ otherwise. The label-conditional Z law alone changes under this shift because the group mixture within each Z changes.

Oracle density control. Oracle calibrates the negative log true conditional density relative to $\mu = \frac{1}{2}U_{K-1} + \frac{1}{2}U_{K-2}$, where the second distribution is uniform over codimension-one faces and uniform within each face. The deletion mechanism yields an analytic Dirichlet face density (Appendix B.1). Its population level sets minimize expected μ-size subject to marginal coverage, illustrating an efficiency benchmark with full distribution knowledge. This is an established density-score construction, not a deployable estimated-density method or a promise of conditional coverage.

Replication and uncertainty. For every setting and $K \in \{3, 8, 20\}$ we run 100 independent repetitions, each with 1,000 pilot, 499 calibration, and 2,000 test examples. Pilot data estimate only

Table 1: Observed joint coverage and normalized size for $K = 3$. Means $\pm$ 95% interval half-width across 100 repetitions. V_2 measures interior area and V_1 measures edge length. H-Z and H-GZ are exploratory response-stratified controls. A displayed zero is rounded, not a claim of exact zero.

	Dense			Zeros ($\pi = .15$)		
Method	Coverage	V_2	V_1	Coverage	V_2	V_1
Max	0.899 ± 0.003	0.074 ± 0.001	0.051 ± 0.001	0.901 ± 0.003	0.149 ± 0.004	0.123 ± 0.004
L2	0.898 ± 0.003	0.074 ± 0.001	0.053 ± 0.001	0.901 ± 0.003	0.151 ± 0.004	0.128 ± 0.005
H	0.899 ± 0.003	0.067 ± 0.001	0.000 ± 0.000	0.901 ± 0.003	0.309 ± 0.012	0.143 ± 0.009
CLR	0.902 ± 0.003	0.134 ± 0.003	0.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
$\text{CLR}_{10^{-3}}$	0.902 ± 0.003	0.129 ± 0.002	0.000 ± 0.000	0.901 ± 0.003	0.941 ± 0.004	0.240 ± 0.010
H-group	0.901 ± 0.003	0.067 ± 0.001	0.000 ± 0.000	0.903 ± 0.003	0.317 ± 0.011	0.152 ± 0.009
H-Z	0.899 ± 0.003	0.067 ± 0.001	1.000 ± 0.000	0.902 ± 0.003	0.067 ± 0.001	0.744 ± 0.004
H-GZ	0.901 ± 0.003	0.067 ± 0.001	1.000 ± 0.000	0.903 ± 0.003	0.067 ± 0.001	0.750 ± 0.004
Oracle	0.898 ± 0.003	0.061 ± 0.001	0.000 ± 0.000	0.901 ± 0.003	0.102 ± 0.001	0.070 ± 0.003

the RMS scales; no threshold or hyperparameter uses test labels. SeedSequence streams derived from master seed 20261102 separate pilot, calibration, test, and integration randomness. Methods share all these draws within each repetition. Tables report means $\pm$ the half-width of a two-sided 95% Student t interval over the 100 repetition-level estimates; method contrasts use paired differences. These are pointwise Monte Carlo intervals for averages over new calibration samples, not simultaneous bands or guarantees for a particular fitted region. Full seed-level distributions are supplied.

For $K = 3$ we estimate Eq. (3) using 2^{14} randomized Sobol joint samples over test covariates and uniform candidate points per repetition, sharing samples across methods. Higher-dimensional size is deliberately not estimated by unreliable rare-hit Monte Carlo. A 2^{18} check on 40 fixed cells changes mean volume estimates by at most 0.0008; an analytic box/simplex calculation independently checks Max (Appendix C). Seven substantive unit checks pass. The final main experiment, including the atom sweep, took 15.8 seconds with four CPU workers on an Intel Xeon Gold 6230; per-process child peak RSS was 163.7 MiB. No GPU was used. All raw data, scripts, timings, and earlier runs are included.

6 Results

6.1 Geometry changes efficiency, not the basic rank guarantee

Dense results illustrate the limited meaning of comparing coverage alone (Table 1). Geometric methods remain close to nominal. H attains coverage 0.899 ± 0.003 and smaller interior area than raw CLR: the paired V_2(H)$-V_2$(CLR) contrast is -0.068 ± 0.002. This is an empirical ranking under the chosen Dirichlet law and interior measure, not a universal ordering of metrics. For a homogeneous interior response, assigning size to unsupported edges penalizes some absolute-error regions as well.

Coordinatewise validity is insufficient: in Dense at $K = 20$, Coord joint coverage is 0.079 ± 0.002, whereas Bonf achieves 0.918 ± 0.003. Dependence imposed by summing to one does not turn separately calibrated coordinate intervals into a jointly valid region. Bonf is a useful conservative baseline but generally cannot allocate error as efficiently as a single joint score. Appendix tables report every method, including both controls.

In Zeros, raw CLR collapses in 100/100 (95% Wilson interval [0.963, 1.000]). Smoothed CLR avoids infinite thresholds but can be very large: V_2 is 0.941 ± 0.004 at $\epsilon = 10^{-3}$, compared with 0.612 ± 0.015 at 10^{-2}. The paired interior-volume difference between H and $\text{CLR}_{10^{-3}}$ is -0.631 ± 0.009. These size statements must be read together with boundary reliability rather than treating all nominally valid regions as interchangeable.

6.2 Boundary and group reliability are separate objectives

In Heterogeneous, pooled H covers the high-dispersion group at 0.588 ± 0.010; H-group reaches 0.900 ± 0.007. Its paired $V_{1/2}$ size difference relative to H is -0.019 ± 0.002: group balancing can improve conditional reliability while also reducing size by avoiding over-inflation for the majority.

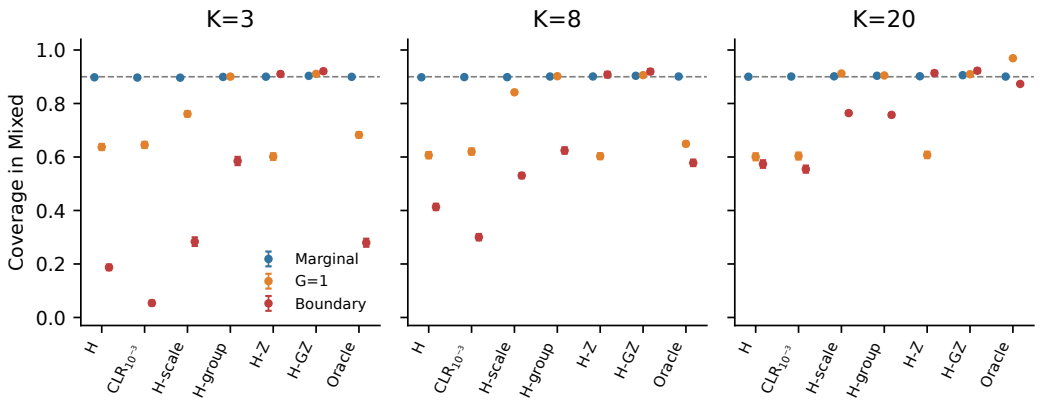


Figure 2: Mixed: marginal, high-dispersion-group, and boundary coverage differ markedly. Bars are pointwise 95% intervals over repetitions. H-group targets G; H-Z targets zero status; H-GZ targets their intersection. Neither a density oracle nor smoothing promises these conditional targets.

Table 2: Reliability diagnostics at $K = 8$. All entries are mean coverage $\pm$ 95% interval half-width. H-weight equals H in the exchangeable Mixed setting; under Shift it uses exact density ratios. The oracle density is not weighted under shift.

	Mixed, $K = 8$			Shift, $K = 8$	
Method	Marginal	$G = 1$	Boundary	Marginal	$G = 1$
H	0.898 ± 0.003	0.607 ± 0.012	0.413 ± 0.012	0.751 ± 0.007	0.619 ± 0.011
$\text{CLR}_{10^{-3}}$	0.899 ± 0.003	0.620 ± 0.012	0.300 ± 0.012	0.756 ± 0.007	0.629 ± 0.010
H-scale	0.899 ± 0.003	0.842 ± 0.006	0.531 ± 0.010	0.870 ± 0.003	0.843 ± 0.005
H-group	0.901 ± 0.003	0.902 ± 0.006	0.624 ± 0.012	0.904 ± 0.003	0.905 ± 0.005
H-Z	0.901 ± 0.003	0.603 ± 0.011	0.908 ± 0.011	0.741 ± 0.007	0.601 ± 0.011
H-GZ	0.904 ± 0.003	0.906 ± 0.006	0.919 ± 0.010	0.906 ± 0.003	0.907 ± 0.005
Oracle	0.901 ± 0.003	0.649 ± 0.009	0.578 ± 0.012	0.761 ± 0.006	0.641 ± 0.008
H-weight	0.898 ± 0.003	0.607 ± 0.012	0.413 ± 0.012	0.903 ± 0.004	0.851 ± 0.006

Oracle still covers $G = 1$ at only 0.675 ± 0.008. Even the true conditional density can allocate errors unevenly when the objective is marginal coverage and average size.

Zeros isolates the support issue without group dispersion. Pooled H's boundary coverage is 0.343 ± 0.017, while H-Z reaches 0.905 ± 0.007. An approximately 90% marginal region can spend much of its allowed error on the 15% boundary subpopulation. Group balancing cannot repair this homogeneous within-group support failure because G is unrelated to it.

Mixed combines both sources of heterogeneity (Figure 2 and Table 2). At $K = 3$, H's boundary coverage is 0.187 ± 0.011, and H-group improves it only to 0.585 ± 0.015. H-scale's $G = 1$ coverage, 0.761 ± 0.010, shows that a pilot RMS correction need not normalize the score tails. H-Z improves boundary coverage to 0.910 ± 0.008 but can retain large group disparities; H-GZ reaches 0.921 ± 0.007 on the boundary and 0.911 ± 0.005 for $G = 1$. The paired boundary-coverage gain of H-GZ relative to H is 0.733 ± 0.012. These are expected consequences of their distinct target guarantees, not evidence that one construction dominates every other criterion.

The size cost can be substantial. Mixed $K = 3$ gives $V_{1/2}$ $=0.490 \pm 0.009$for H-GZ versus 0.178 ± 0.010 for H-group (paired difference 0.312 ± 0.013). The low-zero-probability $G = 0, Z = 1$ cell has mean calibration count 7.25; its threshold is infinite in 71 of 100 repetitions. That correctly includes the entire boundary stratum for those queries. Pooling faces stabilizes boundary calibration but does not eliminate scarcity within the cross-partition.

Oracle attains $V_{1/2}$ $=0.038 \pm 0.001$, but this small size is coupled to weak conditional coverage. Comparing it to H-GZ solely by size would compare different reliability objectives. A reference measure also matters: Appendix Figure 5 varies its interior/edge weight and retains the separate components in all raw outputs.

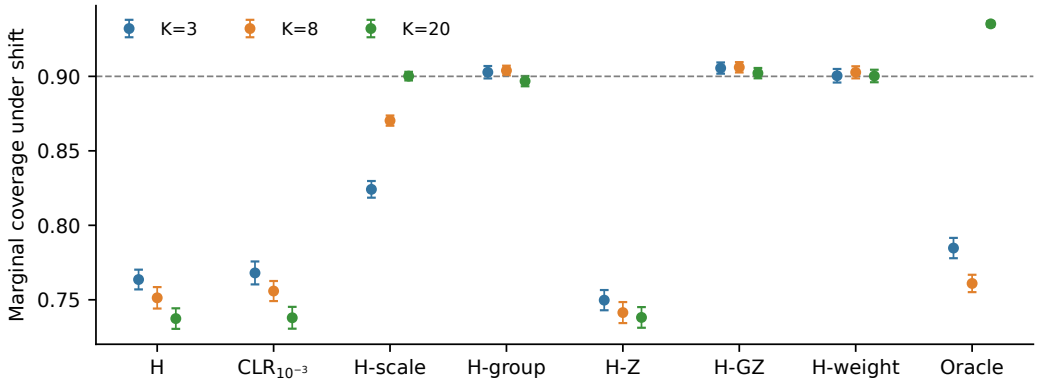


Figure 3: Changing test $P(G = 1)$ from .25 to .65 breaks pooled marginal validity. Exact weighting repairs the marginal target; within-G and within-(G, Z) exchangeability preserve stratified validity. H-Z alone is not invariant to this shift. Intervals quantify variability over independent repetitions.

6.3 Raw and smoothed log ratios fail in different ways

The independent score experiment in Figure 1 agrees with the exact binomial transition and coverage expressions. Raw CLR in Mixed $K = 3$ collapses in 34/100 (95% Wilson interval [0.255, 0.437]), consistent with a zero probability close to the error budget. Both a finite radius that entirely excludes boundary compositions and an infinite radius that includes everything can occur across calibrations. Their average may look reassuring while neither output is attractive.

Smoothing changes this mechanism but supplies no support-conditional guarantee. Mixed $K = 3$ boundary coverage at $\epsilon = 10^{-3}$ is only 0.054 ± 0.009; at 10^{-2} it is 0.134 ± 0.009. Thus the finite-score repair should not be described as a repair of boundary reliability. Appendix Figure 4 reports the full declared sensitivity range, with uncertainty. We select no best ϵ after observing results. The exact zeros here are structural; data rounded to zero, censored measurements, or finite annotation counts would require a separate observation model.

6.4 Shift correction repairs its specified target

Under Shift, H's marginal coverage at $K = 3$ falls to 0.764 ± 0.007. Exact weighting restores it to 0.900 ± 0.005, and H-group obtains 0.903 ± 0.004. At $K = 8$ the paired improvement from weighting H is 0.151 ± 0.006. Weighting does not equalize groups: $K = 3$ H-weight covers $G = 1$ at 0.851 ± 0.007. H-Z alone loses its boundary-conditional exchangeability when the group composition within Z changes; H-GZ retains it because the conditional law given (G, Z) is unchanged. These outcomes verify the assumptions behind the methods rather than suggesting robustness to arbitrary shift. In particular, an unweighted oracle density score does not retain marginal validity merely by knowing the true conditional response law.

7 Discussion and limitations

Our results support a conditional recommendation. For an interior response with an adequate center, standard joint scores are valid and their efficiency should be compared under the geometry relevant to the decision. With exact zeros, inspect how the score defines membership on every relevant face and report support-specific coverage. Raw CLR with the specified infinity extension has a predictable saturation risk; smoothing offers finite scores but should be assessed as a different geometry. Use prespecified covariate or candidate-response stratification when the corresponding conditional coverage is required, budgeting calibration samples for the smallest cells. Under known covariate shift, correct weights address marginal validity; they do not promise conditional equality.

The evidence has deliberate limits. All datasets are synthetic and use a fixed analytic center; we do not establish performance for learned softmax models, topic models, annotation distributions, or empirical latent probabilities. We vary dimension and noise but only use one softmax family, a

Dirichlet interior law, and single-coordinate deletion. We do not study vertices, multiple-zero patterns, continuous sparsity, dependence, estimation of shift weights, or model training instability. The fixed absolute smoothing parameter also has a different relative scale as K increases. Size evidence is restricted to $K = 3$; coverage at larger K does not validate high-dimensional volume rankings. Face size is geometrical and may poorly represent task-specific error costs.

The boundary-stratified controls were added after the initial experiment; their comparisons are exploratory and do not constitute independent confirmation on a new benchmark. Confidence intervals are pointwise, capture Monte Carlo uncertainty under the declared mechanisms, and cannot justify universal method selection. The density oracle uses unavailable information and is a benchmark, not an implementable recommendation. Within a reliability objective, actual applications should select geometry and support partitions before evaluating on held-out targets, and should explicitly define whether the target is an observed composition or an unknown latent distribution.

Broader impact and research process. Support-specific diagnostics can expose harmful overconfidence when zero fractions correspond to consequential outcomes. Conversely, nominal marginal coverage, arbitrary size weights, or overly fine partitions could create a false sense of safety or excessive conservatism. Our artificial groups represent no protected population and do not establish fairness. No personal data or human participants were used. This study was autonomously designed, coded, analyzed, and drafted by an LLM-based research agent; no human review is represented. All reported experiments are executable CPU computations without model API calls, and the complete access and failure log is provided.

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A Proofs and precise targets

Split rank lemma. Conditional on an independently fitted score, assign independent continuous tie-breaking marks to $n+1$ exchangeable scores on the extended real line. The rank of the test score in the resulting total order is uniform. Rank at most k implies numerical inclusion under the nonrandomized threshold in Eq. (1). This establishes coverage at least $k/(n+1) \geq 1-\alpha$; for $k = n+1$ the set is the full candidate domain. Removing the tie-breaking marks can only increase inclusion. This standard argument applies to the arbitrary scores used here and does not assert an upper bound when numerical ties occur.

Proof of Proposition 2. Let $B_n = \sum_{i=1}^{n} \mathbf{1}\{S_i = +\infty\}$. There are $n - B_n$ finite calibration scores. Its kth order statistic is infinite if and only if $n - B_n < k$, which is $B_n > n - k$. Independence gives $B_n \sim \mathrm{Bin}(n, \pi)$, proving Eq. (4).

For the coverage identity, let $N = n+1$ and condition on the unordered N-score sample. With probability one all finite scores are distinct. Write $F = N - B_N$ for their count. By exchangeability, the expected coverage equals the average of the N inclusion events obtained by treating each score in turn as test and calibrating on the other n scores. If $F \geq k$, a finite test score of rank $r \leq k$ is covered: if there are enough remaining finite scores its rank lies below their kth score, and if not the threshold is infinite. A finite test score with $r > k$ is not covered because the remaining calibration sample contains k strictly smaller finite scores. An infinite test score is not covered because the calibration still contains at least k finite scores. Exactly k test choices are therefore covered, giving average k/N.

If $F < k$, removing any score leaves fewer than k finite calibration scores. Every leave-one-out threshold is infinite, so all N choices are included, including the infinite scores under the specified extended-real comparison. Hence unconditional coverage is

$$\frac{k}{N}\mathbb{P}(F \geq k) + \mathbb{P}(F < k) = \frac{k}{N} + \left(1 - \frac{k}{N}\right)\mathbb{P}(B_N > N - k), \tag{6}$$

which is Eq. (5). A finite raw CLR radius cannot include a boundary point; consequently the probability of inclusion for any fixed finite-radius rule is at most $\mathbb{P}(Y \notin \partial\Delta^{K-1}) = 1 - \pi$. This last claim does not bound a random rule that sometimes has infinite radius. The asymptotic transition follows from the law of large numbers for $\pi \neq \alpha$ and the central limit theorem with $k = \lceil N(1-\alpha) \rceil$ at equality.

Response-conditional calibration. Let $A(X, Y)$ be a fixed finite-valued partition, here $Z(Y)$ or $(G(X), Z(Y))$. Conditional on a cell's calibration count n_a and on the test point being in cell a, its calibration pairs and the test pair are i.i.d. from the same cell-conditional distribution whenever that distribution is unchanged across calibration and testing. Applying the split rank lemma gives

$$\mathbb{P}\{s_H(X, Y) \leq q_a \mid A(X, Y) = a, n_a\} \geq 1 - \alpha, \tag{7}$$

with the full stratum included if the rank exceeds n_a. Averaging over n_a preserves the inequality. Region membership in Eq. (2) selects precisely that threshold for the true candidate, establishing the claim. This does not condition on a particular calibration draw or yield pointwise-X coverage. For H-group use $A = G$; for H-GZ use the cross-partition. Under our shift, $Y \mid (T, G)$ and $T \mid G$ are unchanged. Therefore laws within G and within (G, Z) are unchanged, whereas the law within Z alone need not be.

Volume blindness. For fixed x, the indicator difference between $C(x)$ and $C(x) \setminus \partial\Delta^{K-1}$ is supported on the boundary, which has zero $(K-1)$-dimensional Lebesgue measure on the affine hyperplane. Their normalized interior sizes coincide. Their coverage indicator difference is $\mathbf{1}\{Y \in C(X) \cap \partial\Delta^{K-1}\}$, giving Proposition 1 after expectation. The statement does not require any smoothness of C or of the response distribution.

B Data generation, oracle density, and reproducibility

For $j = 0, \ldots, K-1$, define $\theta_j = 2\pi j/K$ and

$$\ell_j(t) = 1.1\cos\theta_j + 1.2t\sin\theta_j + .4t^2\cos(2\theta_j), \qquad m_j(t) = \frac{\exp \ell_j(t)}{\sum_r \exp \ell_r(t)}. \tag{8}$$

The generator draws independent gamma variables with shapes $\kappa_G m_j(t)$ and unit scale and normalizes their sum. It then performs the disclosed zero/deletion step. Every run asserts nonnegativity, finiteness, and that exact zeros arise only from this step, preventing numerical gamma underflow from being mistaken for structural sparsity. Features and target draws are independent across pilot, calibration, and test sets. There is no external dataset or license dependency for the generated observations. Code is released under MIT and generated data under CC0; library versions and notices appear in the accompanying environment files.

H-scale uses $a_g = \sqrt{N_g^{-1} \sum_{i \in \text{pilot}, G_i = g} s_H(X_i, Y_i)^2}$. We do not tune this estimator, the concentrations, the smoothing grid, or the size weight using the test data. Calibration sets for every method are identical. The response strata use exact binary zero status, not an estimated near-zero tolerance. This distinction matters for real measurements.

B.1 Density relative to a stratified measure

Write $a_j = \kappa_g m_j(t)$, and let $\mathrm{Dir}_d(\cdot; a)$ denote the ordinary density in the first $d-1$ coordinates. Uniform normalized simplex density is $\Gamma(d)$ in the same coordinates. Dirichlet aggregation shows that deleting coordinate j and normalizing leaves a Dirichlet vector with parameters a_{-j}, *without* rescaling their total to κ_g. Each deleted coordinate has probability π_g/K.

For an interior point and a point on the relative interior of face j, respectively, the conditional density relative to $\mu = \frac{1}{2}U_{K-1} + \frac{1}{2}U_{K-2}$ is

$$f_\mu(y \mid t, g) = 2(1 - \pi_g) \frac{\mathrm{Dir}_K(y; a)}{\Gamma(K)}, \qquad y \in \operatorname{int} \Delta^{K-1}, \tag{9}$$

$$f_\mu(y \mid t, g) = 2\pi_g \frac{\mathrm{Dir}_{K-1}(y_{-j}; a_{-j})}{\Gamma(K-1)}, \qquad y_j = 0,\ y_{-j} > 0. \tag{10}$$

The factors $1/K$ cancel between actual face probability and reference face-mixture weight. Lower faces carry zero probability under this DGP and this measure; the code sets their density to zero, an arbitrary harmless convention for the reported population metrics. Oracle calibrates $-\log f_\mu$. In no-zero settings it assigns infinite score to face candidates. Its behavior on faces is therefore well defined, unlike using an interior density formula alone at $y_j = 0$.

Population efficiency target. Let $\nu = P_X \otimes \mu$ and $f = dP_{X,Y}/d\nu$. A level set $C_c = \{(x, y) : f(y \mid x) \geq c\}$ has minimal ν-size among measurable sets with at least its coverage, for $c > 0$. Indeed, $(\mathbf{1}_{C_c} - \mathbf{1}_A)(f - c) \geq 0$ pointwise, so

$$P(C_c) - P(A) \geq c\{\nu(C_c) - \nu(A)\}. \tag{11}$$

If $P(A) \geq P(C_c)$, this implies $\nu(A) \geq \nu(C_c)$. A possible density tie can be partially included to match a target. This is a population statement for a specified P_X, reference measure, and coverage, not an exact optimality assertion for randomly calibrated finite-sample regions. Ordinary calibration of this score still needs calibration/test exchangeability, explaining its failure in Shift. Conditional reliability would impose additional constraints and hence a different optimization problem.

B.2 Uncertainty, seeds, and generated outputs

For each metric let M_r be its estimate in repetition r. The displayed half-width is $t_{99,.975}\, \mathrm{sd}(M_1, \ldots, M_{100})/\sqrt{100}$. Conditional coverages first divide by the number of test targets in that stratum within each repetition. Every nonempty response stratum has positive probability in the scenarios where it is reported; no boundary metric is defined in Dense or Heterogeneous. Size integration randomness is also independent between repetitions. Constant empirical metrics can give zero-width t intervals; they should not be interpreted as a universal bound on an unobserved rare event. Collapse proportions in the text instead use Wilson binomial intervals. Paired contrasts are stored with unrounded interval limits in `results/paired_contrasts.csv`. We do not perform multiple-testing-adjusted winner selection.

All seed streams are derived by NumPy SeedSequence from $[20261102, \text{scenario index}, K, r]$, with four children in pilot/calibration/test/integration order. Scenario order is Dense, Heterogeneous, Zeros, Mixed, Shift and $r = 0, \ldots, 99$. The atom sweep uses seed 20261103 and 4,000 actual score arrays

per (n, π), for $n \in \{99, 499, 1999\}$ and the complete probability grid in `src/run_experiments.py`. Conditional test coverage for a realized threshold q is analytically integrated as $(1-\pi)(1-e^{-q})$ for finite q and one otherwise. This reduces test noise without replacing simulated calibration by the exact answer.

Run `bash reproduce.sh` from the artifact root after installing the pinned dependencies. This regenerates experiments, all figures and tables, numerical checks, and the compiled paper. `README.md` documents a results-only build, exact environment commands, expected runtime, and the script for every figure/table. The initial main run is retained in `results/initial`; the unchanged methods agree exactly with the final run on all common reported metrics. A separate two-repetition smoke run and its different method inventory are retained and are not included in paper estimates.

C Numerical validation and sensitivity

For 10 fixed repetitions of each of the four exchangeable $K = 3$ settings, we repeat volume integration with 2^{18} Sobol points using the same scramble. The maximum absolute change in a mean over these 10 repetitions, across scenario/method/size-component combinations, is 0.0008; the maximum change in an individual estimate is 0.0032. These checks took 43.6 seconds and are provided in `results/integration_check.csv`. They validate numerical stability only for these cells, not a uniform integration error bound.

For an independent analytic check, a box $l_j \le y_j \le u_j$ intersected with the K-component simplex has normalized interior volume

$$\sum_{A \subseteq \{1,\dots,K\}} (-1)^{|A|} \left[1 - \sum_j l_j - \sum_{j \in A} (u_j - l_j)\right]_+^{K-1}, \tag{12}$$

obtained by translation and inclusion–exclusion on the upper bounds. For Max set $l_j = \max(0, m_j - q)$ and $u_j = \min(1, m_j + q)$. Numerical quadrature over T evaluates expected volume without sampling candidate compositions. For one calibration of each exchangeable setting, the largest difference from 2^{18}-point Sobol integration is 0.00018. This also checks the normalized volume convention. Unit checks independently verify the oracle density against SciPy's Dirichlet density and exhaustively check the atom coverage identity over leave-one-out choices in a small sample.

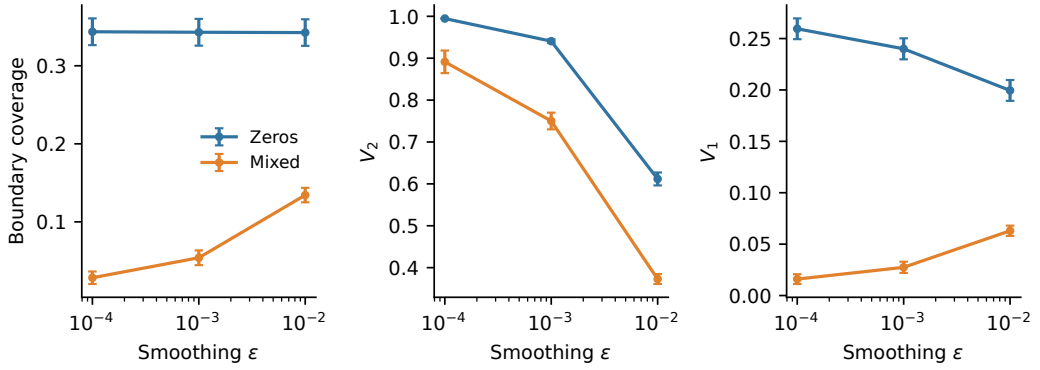


Figure 4: Declared smoothing ablation at $K = 3$. Every point recalibrates its own score on the same calibration data. Larger smoothing changes both size components and can change boundary coverage while preserving only the marginal rank guarantee. All error bars are 95% intervals over repetitions.

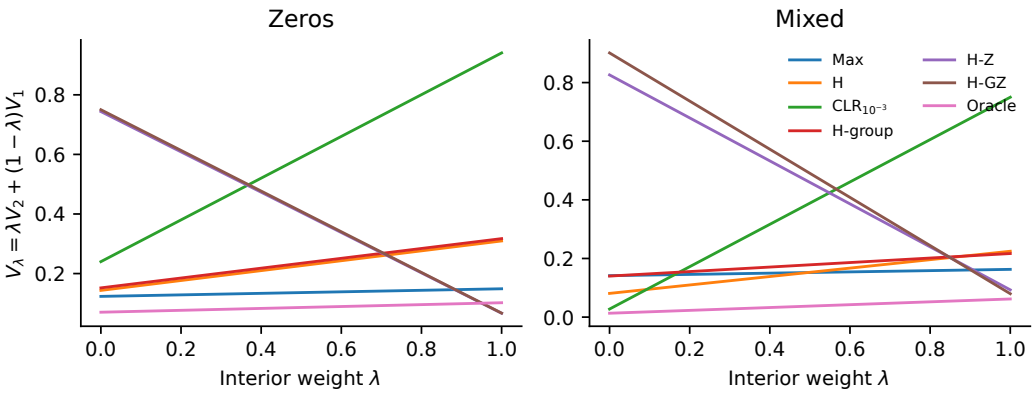


Figure 5: Sensitivity to the reference-measure preference at $K = 3$. Lines combine the actual measured V_2, V_1 means; they are not additional experiments. Each point is an explicit mixture preference, not an empirically identified universal notion of size. Uncertainty for the two components is in the raw summaries.

D Complete results

All entries are means ± pointwise 95% interval half-width over 100 independent repetitions. A dash denotes an undefined or intentionally unestimated metric. $V_{1/2}$ is estimated only for $K = 3$. No boundary targets occur in Dense or Heterogeneous. H-Z/H-GZ are exploratory additions. Tables are generated directly from `results/runs.csv`; unrounded results, seed-level variability, and both size components are in the accompanying CSV files.

D.1 Dense

Method	Marginal	$G = 0$	$G = 1$	Boundary	$V_{1/2}$
$K = 3$					
Max	0.899 ± 0.003	0.899 ± 0.003	0.898 ± 0.004	–	0.063 ± 0.001
L2	0.898 ± 0.003	0.898 ± 0.003	0.897 ± 0.004	–	0.063 ± 0.001
H	0.899 ± 0.003	0.899 ± 0.003	0.899 ± 0.004	–	0.033 ± 0.000
CLR	0.902 ± 0.003	0.902 ± 0.003	0.902 ± 0.004	–	0.067 ± 0.001
$\text{CLR}_{10^{-3}}$	0.902 ± 0.003	0.902 ± 0.003	0.902 ± 0.004	–	0.065 ± 0.001
H-scale	0.899 ± 0.003	0.900 ± 0.003	0.897 ± 0.005	–	0.033 ± 0.000
H-group	0.901 ± 0.003	0.901 ± 0.003	0.902 ± 0.006	–	0.034 ± 0.000
H-Z	0.899 ± 0.003	0.899 ± 0.003	0.899 ± 0.004	–	0.533 ± 0.000
H-GZ	0.901 ± 0.003	0.901 ± 0.003	0.902 ± 0.006	–	0.534 ± 0.000
Oracle	0.898 ± 0.003	0.898 ± 0.003	0.898 ± 0.005	–	0.030 ± 0.000
Coord	0.784 ± 0.004	0.784 ± 0.004	0.784 ± 0.006	–	0.039 ± 0.001
Bonf	0.925 ± 0.003	0.925 ± 0.003	0.925 ± 0.003	–	0.069 ± 0.001
H-weight	0.899 ± 0.003	0.899 ± 0.003	0.899 ± 0.004	–	0.033 ± 0.000
$\text{CLR}_{10^{-2}}$	0.902 ± 0.003	0.902 ± 0.003	0.902 ± 0.004	–	0.050 ± 0.001
$\text{CLR}_{10^{-4}}$	0.902 ± 0.003	0.902 ± 0.003	0.902 ± 0.004	–	0.067 ± 0.001
$K = 8$					
Max	0.897 ± 0.003	0.897 ± 0.003	0.899 ± 0.004	–	–
L2	0.898 ± 0.003	0.897 ± 0.003	0.900 ± 0.004	–	–
H	0.901 ± 0.003	0.901 ± 0.003	0.901 ± 0.004	–	–
CLR	0.900 ± 0.003	0.900 ± 0.003	0.901 ± 0.004	–	–
$\text{CLR}_{10^{-3}}$	0.900 ± 0.003	0.900 ± 0.003	0.902 ± 0.004	–	–
H-scale	0.901 ± 0.003	0.901 ± 0.003	0.899 ± 0.004	–	–
H-group	0.903 ± 0.003	0.902 ± 0.003	0.906 ± 0.006	–	–
H-Z	0.901 ± 0.003	0.901 ± 0.003	0.901 ± 0.004	–	–
H-GZ	0.903 ± 0.003	0.902 ± 0.003	0.906 ± 0.006	–	–
Oracle	0.901 ± 0.003	0.901 ± 0.003	0.901 ± 0.004	–	–
Coord	0.423 ± 0.005	0.423 ± 0.005	0.424 ± 0.006	–	–
Bonf	0.906 ± 0.003	0.906 ± 0.003	0.908 ± 0.004	–	–
H-weight	0.901 ± 0.003	0.901 ± 0.003	0.901 ± 0.004	–	–
$\text{CLR}_{10^{-2}}$	0.900 ± 0.003	0.901 ± 0.003	0.900 ± 0.004	–	–
$\text{CLR}_{10^{-4}}$	0.900 ± 0.003	0.900 ± 0.003	0.901 ± 0.004	–	–
$K = 20$					
Max	0.896 ± 0.003	0.897 ± 0.003	0.896 ± 0.004	–	–
L2	0.897 ± 0.003	0.898 ± 0.003	0.896 ± 0.003	–	–
H	0.898 ± 0.003	0.899 ± 0.003	0.896 ± 0.004	–	–
CLR	0.899 ± 0.003	0.900 ± 0.003	0.896 ± 0.004	–	–
$\text{CLR}_{10^{-3}}$	0.900 ± 0.003	0.899 ± 0.003	0.901 ± 0.004	–	–
H-scale	0.898 ± 0.003	0.899 ± 0.003	0.896 ± 0.004	–	–
H-group	0.899 ± 0.003	0.899 ± 0.003	0.900 ± 0.006	–	–
H-Z	0.898 ± 0.003	0.899 ± 0.003	0.896 ± 0.004	–	–
H-GZ	0.899 ± 0.003	0.899 ± 0.003	0.900 ± 0.006	–	–
Oracle	0.900 ± 0.003	0.899 ± 0.003	0.903 ± 0.004	–	–
Coord	0.079 ± 0.002	0.079 ± 0.002	0.078 ± 0.003	–	–
Bonf	0.918 ± 0.003	0.919 ± 0.003	0.917 ± 0.004	–	–
H-weight	0.898 ± 0.003	0.899 ± 0.003	0.896 ± 0.004	–	–
$\text{CLR}_{10^{-2}}$	0.896 ± 0.004	0.896 ± 0.004	0.896 ± 0.004	–	–
$\text{CLR}_{10^{-4}}$	0.900 ± 0.003	0.901 ± 0.003	0.900 ± 0.004	–	–

D.2 Heterogeneous

Method	Marginal	$G = 0$	$G = 1$	Boundary	$V_{1/2}$
$K = 3$					
Max	0.898 ± 0.003	1.000 ± 0.000	0.594 ± 0.010	–	0.093 ± 0.002
L2	0.898 ± 0.003	1.000 ± 0.000	0.594 ± 0.010	–	0.093 ± 0.002
H	0.897 ± 0.003	1.000 ± 0.000	0.588 ± 0.010	–	0.070 ± 0.002
CLR	0.896 ± 0.003	1.000 ± 0.000	0.586 ± 0.010	–	0.130 ± 0.005
$\text{CLR}_{10^{-3}}$	0.896 ± 0.003	1.000 ± 0.000	0.586 ± 0.010	–	0.126 ± 0.004
H-scale	0.900 ± 0.003	0.895 ± 0.003	0.914 ± 0.005	–	0.053 ± 0.001
H-group	0.901 ± 0.003	0.901 ± 0.003	0.900 ± 0.007	–	0.051 ± 0.001
H-Z	0.897 ± 0.003	1.000 ± 0.000	0.588 ± 0.010	–	0.561 ± 0.001
H-GZ	0.901 ± 0.003	0.901 ± 0.003	0.900 ± 0.007	–	0.538 ± 0.001
Oracle	0.898 ± 0.003	0.972 ± 0.001	0.675 ± 0.008	–	0.019 ± 0.000
Coord	0.824 ± 0.003	0.989 ± 0.001	0.329 ± 0.009	–	0.039 ± 0.001
Bonf	0.930 ± 0.002	1.000 ± 0.000	0.722 ± 0.008	–	0.121 ± 0.002
H-weight	0.897 ± 0.003	1.000 ± 0.000	0.588 ± 0.010	–	0.070 ± 0.002
$\text{CLR}_{10^{-2}}$	0.895 ± 0.003	1.000 ± 0.000	0.581 ± 0.010	–	0.095 ± 0.002
$\text{CLR}_{10^{-4}}$	0.896 ± 0.003	1.000 ± 0.000	0.586 ± 0.010	–	0.130 ± 0.004
$K = 8$					
Max	0.901 ± 0.003	1.000 ± 0.000	0.603 ± 0.011	–	–
L2	0.902 ± 0.003	1.000 ± 0.000	0.608 ± 0.011	–	–
H	0.902 ± 0.003	1.000 ± 0.000	0.604 ± 0.011	–	–
CLR	0.902 ± 0.003	1.000 ± 0.000	0.606 ± 0.011	–	–
$\text{CLR}_{10^{-3}}$	0.902 ± 0.003	1.000 ± 0.000	0.605 ± 0.011	–	–
H-scale	0.900 ± 0.003	0.895 ± 0.003	0.917 ± 0.004	–	–
H-group	0.903 ± 0.003	0.900 ± 0.004	0.910 ± 0.006	–	–
H-Z	0.902 ± 0.003	1.000 ± 0.000	0.604 ± 0.011	–	–
H-GZ	0.903 ± 0.003	0.900 ± 0.004	0.910 ± 0.006	–	–
Oracle	0.898 ± 0.002	0.981 ± 0.001	0.646 ± 0.007	–	–
Coord	0.697 ± 0.003	0.922 ± 0.003	0.020 ± 0.002	–	–
Bonf	0.912 ± 0.002	1.000 ± 0.000	0.644 ± 0.009	–	–
H-weight	0.902 ± 0.003	1.000 ± 0.000	0.604 ± 0.011	–	–
$\text{CLR}_{10^{-2}}$	0.901 ± 0.003	1.000 ± 0.000	0.601 ± 0.011	–	–
$\text{CLR}_{10^{-4}}$	0.901 ± 0.003	1.000 ± 0.000	0.604 ± 0.011	–	–
$K = 20$					
Max	0.899 ± 0.003	1.000 ± 0.000	0.596 ± 0.011	–	–
L2	0.901 ± 0.003	1.000 ± 0.000	0.604 ± 0.011	–	–
H	0.902 ± 0.003	1.000 ± 0.000	0.607 ± 0.011	–	–
CLR	0.902 ± 0.003	1.000 ± 0.000	0.609 ± 0.011	–	–
$\text{CLR}_{10^{-3}}$	0.902 ± 0.003	1.000 ± 0.000	0.607 ± 0.011	–	–
H-scale	0.900 ± 0.003	0.895 ± 0.003	0.915 ± 0.004	–	–
H-group	0.903 ± 0.003	0.903 ± 0.003	0.904 ± 0.006	–	–
H-Z	0.902 ± 0.003	1.000 ± 0.000	0.607 ± 0.011	–	–
H-GZ	0.903 ± 0.003	0.903 ± 0.003	0.904 ± 0.006	–	–
Oracle	0.903 ± 0.003	0.879 ± 0.004	0.973 ± 0.001	–	–
Coord	0.443 ± 0.005	0.590 ± 0.006	0.000 ± 0.000	–	–
Bonf	0.924 ± 0.002	1.000 ± 0.000	0.693 ± 0.009	–	–
H-weight	0.902 ± 0.003	1.000 ± 0.000	0.607 ± 0.011	–	–
$\text{CLR}_{10^{-2}}$	0.902 ± 0.003	1.000 ± 0.000	0.607 ± 0.012	–	–
$\text{CLR}_{10^{-4}}$	0.901 ± 0.003	1.000 ± 0.000	0.602 ± 0.011	–	–

D.3 Zeros

Method	Marginal	$G = 0$	$G = 1$	Boundary	$V_{1/2}$
$K = 3$					
Max	0.901 ± 0.003	0.901 ± 0.003	0.900 ± 0.004	0.410 ± 0.010	0.136 ± 0.004
L2	0.901 ± 0.003	0.902 ± 0.003	0.900 ± 0.004	0.411 ± 0.010	0.140 ± 0.004
H	0.901 ± 0.003	0.902 ± 0.003	0.900 ± 0.004	0.343 ± 0.017	0.226 ± 0.011
CLR	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
$\text{CLR}_{10^{-3}}$	0.901 ± 0.003	0.902 ± 0.003	0.901 ± 0.004	0.343 ± 0.017	0.590 ± 0.007
H-scale	0.901 ± 0.003	0.902 ± 0.003	0.900 ± 0.004	0.343 ± 0.017	0.226 ± 0.010
H-group	0.903 ± 0.003	0.904 ± 0.003	0.900 ± 0.006	0.357 ± 0.017	0.235 ± 0.010
H-Z	0.902 ± 0.003	0.902 ± 0.003	0.901 ± 0.004	0.905 ± 0.007	0.405 ± 0.002
H-GZ	0.903 ± 0.003	0.902 ± 0.003	0.905 ± 0.005	0.913 ± 0.007	0.409 ± 0.002
Oracle	0.901 ± 0.003	0.902 ± 0.003	0.899 ± 0.004	0.459 ± 0.016	0.086 ± 0.002
Coord	0.833 ± 0.004	0.834 ± 0.005	0.831 ± 0.005	0.263 ± 0.007	0.075 ± 0.002
Bonf	0.946 ± 0.002	0.947 ± 0.002	0.945 ± 0.003	0.647 ± 0.010	0.315 ± 0.013
H-weight	0.901 ± 0.003	0.902 ± 0.003	0.900 ± 0.004	0.343 ± 0.017	0.226 ± 0.011
$\text{CLR}_{10^{-2}}$	0.901 ± 0.003	0.902 ± 0.003	0.901 ± 0.004	0.343 ± 0.017	0.406 ± 0.013
$\text{CLR}_{10^{-4}}$	0.901 ± 0.003	0.902 ± 0.003	0.901 ± 0.004	0.344 ± 0.017	0.627 ± 0.005
$K = 8$					
Max	0.900 ± 0.003	0.900 ± 0.003	0.899 ± 0.004	0.639 ± 0.007	–
L2	0.900 ± 0.003	0.900 ± 0.003	0.900 ± 0.003	0.614 ± 0.006	–
H	0.899 ± 0.003	0.900 ± 0.003	0.898 ± 0.004	0.409 ± 0.009	–
CLR	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	–
$\text{CLR}_{10^{-3}}$	0.899 ± 0.003	0.899 ± 0.003	0.899 ± 0.003	0.329 ± 0.015	–
H-scale	0.899 ± 0.003	0.900 ± 0.003	0.898 ± 0.004	0.410 ± 0.009	–
H-group	0.901 ± 0.003	0.901 ± 0.003	0.901 ± 0.006	0.418 ± 0.009	–
H-Z	0.902 ± 0.003	0.902 ± 0.003	0.902 ± 0.004	0.900 ± 0.008	–
H-GZ	0.904 ± 0.003	0.904 ± 0.003	0.904 ± 0.005	0.904 ± 0.007	–
Oracle	0.900 ± 0.003	0.900 ± 0.003	0.901 ± 0.004	0.685 ± 0.010	–
Coord	0.450 ± 0.004	0.449 ± 0.004	0.452 ± 0.006	0.196 ± 0.005	–
Bonf	0.916 ± 0.003	0.916 ± 0.003	0.916 ± 0.003	0.706 ± 0.009	–
H-weight	0.899 ± 0.003	0.900 ± 0.003	0.898 ± 0.004	0.409 ± 0.009	–
$\text{CLR}_{10^{-2}}$	0.900 ± 0.003	0.900 ± 0.003	0.899 ± 0.003	0.384 ± 0.010	–
$\text{CLR}_{10^{-4}}$	0.899 ± 0.003	0.899 ± 0.003	0.899 ± 0.003	0.324 ± 0.016	–
$K = 20$					
Max	0.898 ± 0.003	0.897 ± 0.003	0.899 ± 0.004	0.841 ± 0.006	–
L2	0.900 ± 0.003	0.899 ± 0.003	0.901 ± 0.004	0.791 ± 0.006	–
H	0.904 ± 0.003	0.903 ± 0.003	0.905 ± 0.003	0.621 ± 0.007	–
CLR	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	–
$\text{CLR}_{10^{-3}}$	0.903 ± 0.003	0.903 ± 0.003	0.903 ± 0.004	0.529 ± 0.009	–
H-scale	0.904 ± 0.003	0.904 ± 0.003	0.903 ± 0.004	0.621 ± 0.008	–
H-group	0.905 ± 0.003	0.904 ± 0.003	0.909 ± 0.006	0.625 ± 0.008	–
H-Z	0.904 ± 0.003	0.903 ± 0.003	0.905 ± 0.004	0.907 ± 0.007	–
H-GZ	0.905 ± 0.003	0.903 ± 0.004	0.912 ± 0.006	0.913 ± 0.008	–
Oracle	0.903 ± 0.003	0.903 ± 0.003	0.904 ± 0.004	0.813 ± 0.006	–
Coord	0.083 ± 0.002	0.083 ± 0.002	0.083 ± 0.003	0.050 ± 0.002	–
Bonf	0.921 ± 0.002	0.921 ± 0.002	0.920 ± 0.003	0.894 ± 0.004	–
H-weight	0.904 ± 0.003	0.903 ± 0.003	0.905 ± 0.003	0.621 ± 0.007	–
$\text{CLR}_{10^{-2}}$	0.903 ± 0.003	0.903 ± 0.003	0.904 ± 0.004	0.631 ± 0.007	–
$\text{CLR}_{10^{-4}}$	0.901 ± 0.003	0.900 ± 0.003	0.902 ± 0.004	0.521 ± 0.011	–

D.4 Mixed

Method	Marginal	$G = 0$	$G = 1$	Boundary	$V_{1/2}$
$K = 3$					
Max	0.898 ± 0.003	0.989 ± 0.001	0.624 ± 0.010	0.331 ± 0.010	0.152 ± 0.004
L2	0.898 ± 0.003	0.989 ± 0.001	0.624 ± 0.010	0.329 ± 0.010	0.152 ± 0.004
H	0.898 ± 0.003	0.985 ± 0.001	0.637 ± 0.011	0.187 ± 0.011	0.153 ± 0.006
CLR	0.927 ± 0.011	0.987 ± 0.002	0.749 ± 0.037	0.340 ± 0.094	0.625 ± 0.055
$\text{CLR}_{10^{-3}}$	0.897 ± 0.003	0.981 ± 0.001	0.645 ± 0.012	0.054 ± 0.009	0.389 ± 0.012
H-scale	0.897 ± 0.003	0.942 ± 0.004	0.761 ± 0.010	0.283 ± 0.016	0.074 ± 0.003
H-group	0.899 ± 0.003	0.899 ± 0.003	0.901 ± 0.006	0.585 ± 0.015	0.178 ± 0.010
H-Z	0.900 ± 0.003	1.000 ± 0.000	0.602 ± 0.012	0.910 ± 0.008	0.460 ± 0.004
H-GZ	0.903 ± 0.003	0.901 ± 0.004	0.911 ± 0.005	0.921 ± 0.007	0.490 ± 0.009
Oracle	0.900 ± 0.003	0.972 ± 0.001	0.683 ± 0.010	0.279 ± 0.015	0.038 ± 0.001
Coord	0.828 ± 0.003	0.983 ± 0.001	0.363 ± 0.011	0.154 ± 0.006	0.063 ± 0.002
Bonf	0.942 ± 0.002	0.992 ± 0.001	0.790 ± 0.009	0.566 ± 0.015	0.245 ± 0.009
H-weight	0.898 ± 0.003	0.985 ± 0.001	0.637 ± 0.011	0.187 ± 0.011	0.153 ± 0.006
$\text{CLR}_{10^{-2}}$	0.896 ± 0.003	0.983 ± 0.001	0.638 ± 0.011	0.134 ± 0.009	0.218 ± 0.008
$\text{CLR}_{10^{-4}}$	0.896 ± 0.003	0.981 ± 0.001	0.644 ± 0.011	0.028 ± 0.008	0.454 ± 0.015
$K = 8$					
Max	0.900 ± 0.003	0.995 ± 0.000	0.611 ± 0.011	0.515 ± 0.011	–
L2	0.899 ± 0.003	0.996 ± 0.000	0.604 ± 0.011	0.509 ± 0.012	–
H	0.898 ± 0.003	0.995 ± 0.000	0.607 ± 0.012	0.413 ± 0.012	–
CLR	0.931 ± 0.011	0.987 ± 0.002	0.762 ± 0.037	0.370 ± 0.096	–
$\text{CLR}_{10^{-3}}$	0.899 ± 0.003	0.991 ± 0.001	0.620 ± 0.012	0.300 ± 0.012	–
H-scale	0.899 ± 0.003	0.917 ± 0.003	0.842 ± 0.006	0.531 ± 0.010	–
H-group	0.901 ± 0.003	0.900 ± 0.004	0.902 ± 0.006	0.624 ± 0.012	–
H-Z	0.901 ± 0.003	1.000 ± 0.000	0.603 ± 0.011	0.908 ± 0.011	–
H-GZ	0.904 ± 0.003	0.903 ± 0.004	0.906 ± 0.006	0.919 ± 0.010	–
Oracle	0.901 ± 0.003	0.984 ± 0.001	0.649 ± 0.009	0.578 ± 0.012	–
Coord	0.694 ± 0.003	0.917 ± 0.003	0.022 ± 0.002	0.055 ± 0.004	–
Bonf	0.911 ± 0.002	0.999 ± 0.000	0.647 ± 0.009	0.605 ± 0.011	–
H-weight	0.898 ± 0.003	0.995 ± 0.000	0.607 ± 0.012	0.413 ± 0.012	–
$\text{CLR}_{10^{-2}}$	0.899 ± 0.003	0.992 ± 0.001	0.616 ± 0.012	0.392 ± 0.012	–
$\text{CLR}_{10^{-4}}$	0.898 ± 0.003	0.990 ± 0.001	0.620 ± 0.011	0.244 ± 0.013	–
$K = 20$					
Max	0.899 ± 0.003	1.000 ± 0.000	0.594 ± 0.012	0.633 ± 0.011	–
L2	0.899 ± 0.003	1.000 ± 0.000	0.598 ± 0.012	0.619 ± 0.013	–
H	0.900 ± 0.003	1.000 ± 0.000	0.601 ± 0.013	0.574 ± 0.014	–
CLR	0.936 ± 0.011	0.988 ± 0.002	0.778 ± 0.037	0.400 ± 0.098	–
$\text{CLR}_{10^{-3}}$	0.901 ± 0.003	1.000 ± 0.000	0.604 ± 0.013	0.555 ± 0.013	–
H-scale	0.902 ± 0.003	0.898 ± 0.003	0.912 ± 0.004	0.764 ± 0.007	–
H-group	0.904 ± 0.003	0.903 ± 0.003	0.905 ± 0.005	0.757 ± 0.007	–
H-Z	0.902 ± 0.003	1.000 ± 0.000	0.608 ± 0.012	0.914 ± 0.009	–
H-GZ	0.906 ± 0.003	0.905 ± 0.004	0.909 ± 0.005	0.923 ± 0.008	–
Oracle	0.901 ± 0.003	0.878 ± 0.004	0.969 ± 0.002	0.873 ± 0.004	–
Coord	0.438 ± 0.006	0.584 ± 0.008	0.000 ± 0.000	0.046 ± 0.003	–
Bonf	0.922 ± 0.003	1.000 ± 0.000	0.690 ± 0.010	0.721 ± 0.011	–
H-weight	0.900 ± 0.003	1.000 ± 0.000	0.601 ± 0.013	0.574 ± 0.014	–
$\text{CLR}_{10^{-2}}$	0.902 ± 0.003	1.000 ± 0.000	0.607 ± 0.012	0.604 ± 0.013	–
$\text{CLR}_{10^{-4}}$	0.901 ± 0.003	1.000 ± 0.000	0.605 ± 0.013	0.514 ± 0.013	–

D.5 Shift

Method	Marginal	$G = 0$	$G = 1$	Boundary	$V_{1/2}$
$K = 3$					
Max	0.756 ± 0.006	0.990 ± 0.001	0.630 ± 0.010	0.317 ± 0.010	0.154 ± 0.004
L2	0.757 ± 0.007	0.989 ± 0.001	0.631 ± 0.010	0.318 ± 0.011	0.156 ± 0.005
H	0.764 ± 0.007	0.985 ± 0.001	0.644 ± 0.010	0.180 ± 0.010	0.157 ± 0.006
CLR	0.843 ± 0.025	0.988 ± 0.002	0.764 ± 0.038	0.380 ± 0.097	0.649 ± 0.056
$\text{CLR}_{10^{-3}}$	0.768 ± 0.008	0.981 ± 0.001	0.653 ± 0.012	0.067 ± 0.012	0.399 ± 0.014
H-scale	0.824 ± 0.006	0.946 ± 0.004	0.758 ± 0.009	0.318 ± 0.017	0.171 ± 0.006
H-group	0.903 ± 0.004	0.902 ± 0.004	0.903 ± 0.006	0.679 ± 0.016	0.452 ± 0.027
H-Z	0.750 ± 0.007	1.000 ± 0.000	0.615 ± 0.010	0.900 ± 0.010	0.462 ± 0.004
H-GZ	0.906 ± 0.004	0.905 ± 0.004	0.906 ± 0.005	0.916 ± 0.008	0.520 ± 0.005
Oracle	0.785 ± 0.007	0.972 ± 0.001	0.684 ± 0.010	0.324 ± 0.019	0.069 ± 0.003
Coord	0.588 ± 0.007	0.984 ± 0.001	0.374 ± 0.010	0.141 ± 0.006	0.065 ± 0.002
Bonf	0.861 ± 0.006	0.992 ± 0.001	0.791 ± 0.009	0.557 ± 0.015	0.245 ± 0.009
H-weight	0.900 ± 0.005	0.993 ± 0.001	0.851 ± 0.007	0.553 ± 0.019	0.423 ± 0.032
$\text{CLR}_{10^{-2}}$	0.768 ± 0.007	0.983 ± 0.001	0.651 ± 0.010	0.143 ± 0.010	0.230 ± 0.009
$\text{CLR}_{10^{-4}}$	0.769 ± 0.008	0.981 ± 0.001	0.654 ± 0.012	0.048 ± 0.012	0.464 ± 0.016
$K = 8$					
Max	0.749 ± 0.006	0.995 ± 0.000	0.615 ± 0.010	0.481 ± 0.009	–
L2	0.747 ± 0.006	0.997 ± 0.000	0.612 ± 0.009	0.473 ± 0.009	–
H	0.751 ± 0.007	0.995 ± 0.000	0.619 ± 0.011	0.379 ± 0.011	–
CLR	0.840 ± 0.024	0.987 ± 0.002	0.760 ± 0.036	0.350 ± 0.095	–
$\text{CLR}_{10^{-3}}$	0.756 ± 0.007	0.991 ± 0.001	0.629 ± 0.010	0.271 ± 0.010	–
H-scale	0.870 ± 0.003	0.921 ± 0.003	0.843 ± 0.005	0.608 ± 0.008	–
H-group	0.904 ± 0.003	0.901 ± 0.003	0.905 ± 0.005	0.718 ± 0.010	–
H-Z	0.741 ± 0.007	1.000 ± 0.000	0.601 ± 0.011	0.898 ± 0.008	–
H-GZ	0.906 ± 0.003	0.904 ± 0.004	0.907 ± 0.005	0.912 ± 0.008	–
Oracle	0.761 ± 0.006	0.983 ± 0.002	0.641 ± 0.008	0.559 ± 0.010	–
Coord	0.336 ± 0.002	0.919 ± 0.003	0.021 ± 0.002	0.019 ± 0.002	–
Bonf	0.774 ± 0.005	0.999 ± 0.000	0.653 ± 0.008	0.568 ± 0.010	–
H-weight	0.903 ± 0.004	0.997 ± 0.000	0.851 ± 0.006	0.651 ± 0.010	–
$\text{CLR}_{10^{-2}}$	0.750 ± 0.007	0.992 ± 0.001	0.618 ± 0.010	0.369 ± 0.010	–
$\text{CLR}_{10^{-4}}$	0.759 ± 0.007	0.990 ± 0.001	0.634 ± 0.010	0.223 ± 0.011	–
$K = 20$					
Max	0.736 ± 0.007	1.000 ± 0.000	0.595 ± 0.011	0.576 ± 0.012	–
L2	0.736 ± 0.006	1.000 ± 0.000	0.594 ± 0.010	0.560 ± 0.011	–
H	0.737 ± 0.007	1.000 ± 0.000	0.596 ± 0.011	0.508 ± 0.012	–
CLR	0.863 ± 0.025	0.990 ± 0.002	0.796 ± 0.038	0.460 ± 0.099	–
$\text{CLR}_{10^{-3}}$	0.738 ± 0.007	1.000 ± 0.000	0.597 ± 0.011	0.481 ± 0.012	–
H-scale	0.900 ± 0.003	0.889 ± 0.004	0.906 ± 0.003	0.835 ± 0.005	–
H-group	0.897 ± 0.003	0.894 ± 0.004	0.898 ± 0.005	0.825 ± 0.007	–
H-Z	0.738 ± 0.007	1.000 ± 0.000	0.598 ± 0.010	0.890 ± 0.008	–
H-GZ	0.902 ± 0.003	0.896 ± 0.004	0.905 ± 0.005	0.910 ± 0.007	–
Oracle	0.935 ± 0.002	0.873 ± 0.005	0.969 ± 0.001	0.944 ± 0.002	–
Coord	0.203 ± 0.003	0.581 ± 0.007	0.000 ± 0.000	0.009 ± 0.001	–
Bonf	0.802 ± 0.006	1.000 ± 0.000	0.696 ± 0.010	0.682 ± 0.011	–
H-weight	0.900 ± 0.004	1.000 ± 0.000	0.847 ± 0.006	0.785 ± 0.009	–
$\text{CLR}_{10^{-2}}$	0.740 ± 0.007	1.000 ± 0.000	0.601 ± 0.010	0.533 ± 0.011	–
$\text{CLR}_{10^{-4}}$	0.742 ± 0.008	1.000 ± 0.000	0.604 ± 0.011	0.438 ± 0.012	–

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